Practical - 1

Aim:-
Study of various Network commands used in Linux and windows.

Basic Networking Commands

i) arp - a
purpose:- It will show the IP address of your computer along with the IP address and MAC address of your router
Command : arp -a
o/p : _gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3

ii) host name
purpose: Shows the computer name
command: hostname
o/p : localhost.localdomain

iii) ipconfig
purpose: it shows network configuration details
cmd : ipconfig
o/p : Wireless LAN adapter Wi-Fi :
Connection-specific DNS suffix . :
Link-local IPv6 Address . . . . : fe80::6768:d34:5181:ef2a%4
IPv4 Address . . . . . : 172.16.11.108
Subnet Mask . . . . . : 255.255.252.0
Default Gateway . . . . . : 172.16.8.1

Practical - 1

Aim:-
Study of various Network commands used in Linux and windows.

Basic Networking Commands

i) arp - a
purpose:- It will show the IP address of your computer along with the IP address and MAC address of your router
Command : arp -a
o/p : gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3

ii) host name
purpose: Shows the computer name
command: hostname
o/p : localhost.localdomain

iii) ipconfig
purpose: it Shows network configuration details
cmd : ipconfig
o/p : Wireless LAN adapter Wi-Fi :
Connection-specific DNS suffix . :
Link-local IPv6 Address . . . . : fe80::6768:d34:5181:ef2a%4
IPv4 Address . . . . . : 172.16.11.108
Subnet Mask . . . . . : 255.255.252.0
Default Gateway . . . . . : 172.16.8.1

iv) nbtstat -a

purpose :- Shows NetBIOS information

iv) Command :- nslookup www.google.com

Description :- Tool used to perform DNS lookups in Linux

output :- Non-authoritative answer

Name : www.google.com

Address: 142.251.152.119

v.) Command : pathing

Description : pathing is unique to widow and is basically a combination of the ping and tracert cmds

vi) Command : Ping

Description : Test the connectivity between two nodes.

i.) Ping local host

o/p:

64 bytes from local host (::1): icmp-seq=1

64 bytes from local host (::1): icmp-seq=2 ttl=04 time=0.125 ms

ii) Ping 4.2.2.2

output :

64 bytes from 4.2.2.2 icmp-seq=1 ttl=82 time=255 ms

64 bytes from 4.2.2.2 icmp-seq=2 ttl=52 time=221 ms

iii) ping www.facebook.com

o/p:-

64 bytes from edge-star-mini-shv-02-maas.facebook.com (57.144.56.1): icmp-seq=1 ttl=1 time=35.1 ms

vii) Command : Route

Description : Its used to manipulate the IP routing table

Output :

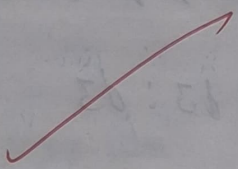
Kernel IP routing table

Destination	Gateway	Genmask	Flag	Metric	Ref	Use
default	- gateway	0.0.0.0	UG	100	0	0
10.0.2.0	0.0.0.0	255.255.255	U	100	0	0

Iface

enpos3

enpos3



Linux Networking Command

1.) Command : IP

Description : It can show address information, manipulate various devices & tunnels

Some Common use case IP commands :

i) ip address show

output :

enpos3 : <BROADCAST, multicast, UP, LOWER-UP>
mtu 1500 qdisc fq_codel state up group default qlen 1000
link/ether 08:00:27:90:d3:d3
brd ff:ff:ff
10.0.2.15/25 brd 10.02.255 scope global dynamic no prefix route enpos3 valid - ft

ii) ip address add 192.168.1.254/24 dev enpos3

o/p :
inet 192.168.1.254/24 scope global enpos3
The address has been added in enpos3.

iii) ip address delete 192.168.1.254/25 dev enpos3

o/p : The ip address 192.168.1.254/25 was successfully deleted from the network interface enpos3

ii) Command : ifconfig

Description :- It is a state in many sysadmin's tool belt for configuring & troubleshooting networks.

output :-

lo: flags = 73<UP, LOOPBACK, RUNNING> MTU 65536
inet 127.0.0.1 netmask 255.0.0.0
inet ::1 prefixlen 128 scopeid 0x10<host>
loop txqueue

iii) Command : mtr

Description : It serves as a network diagnostic & troubleshooting tool.

i) mtr google.com

O/P :-

ii) mtr -g google.com

O/P :
-U, --udp use UPP instead of ICMP echo
-T, --tcp use TCP instead of ICMP echo

iii) mtr -6 google.com

O/P :-

	Host	Loss	Snt	Last	Avg	Best	Worst	
.	gateway	0.0%	9	0.8	1.4	0.5	5.5	1.3

iv) mtr -C 10

O/P :-

Host	Loss %	Snt	Last	Avg	Best	Worst
Localhost	0.0%	2	0.1	0.1	0.1	0.4

output :-

i) dnf install -y tcpdump

O/P :- Dependies resolved
Nothing to do complete

ii) tcpdump -D

o/p :- bluetooth - monitor [wireless]
usbmon2 [Raw USB traffic, bus number 2)

iii) tcp dump -i eth0

o/p

iv) tcpdump -I eth0 -c 10

o/p

v.) tcpdump -i eth0

o/p

Result :-

Thus the network command using linux & window are successfully executed & the o/p is verified.